



REACH CAD

Learning Guide: Adding Internal Details and Advanced
Viewing in REACH CAD

Learning Guide

Learning Guide: Adding Internal Details and Advanced Viewing in REACH CAD Objective

By the end of this lesson, you will be able to:

- Add and manage essential internal lines (grainline, center line, mirror line, balance line) in pattern pieces.
- Use the Convert tool to change line properties.
- Master the View menu to control and optimize your workspace display for complex styles.

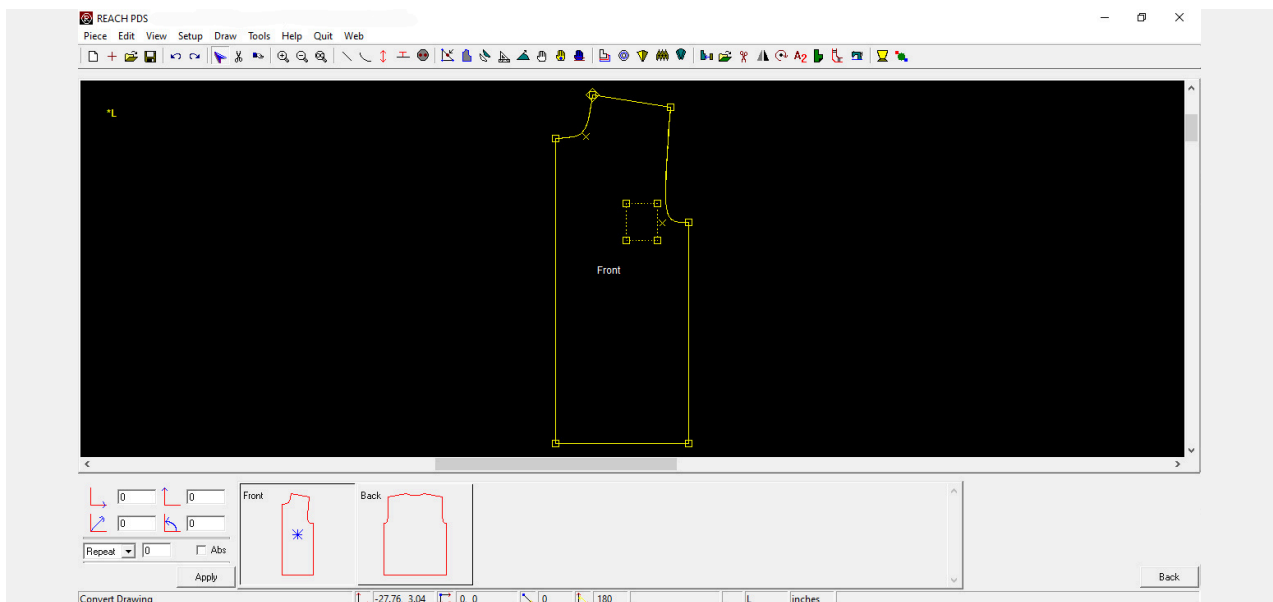
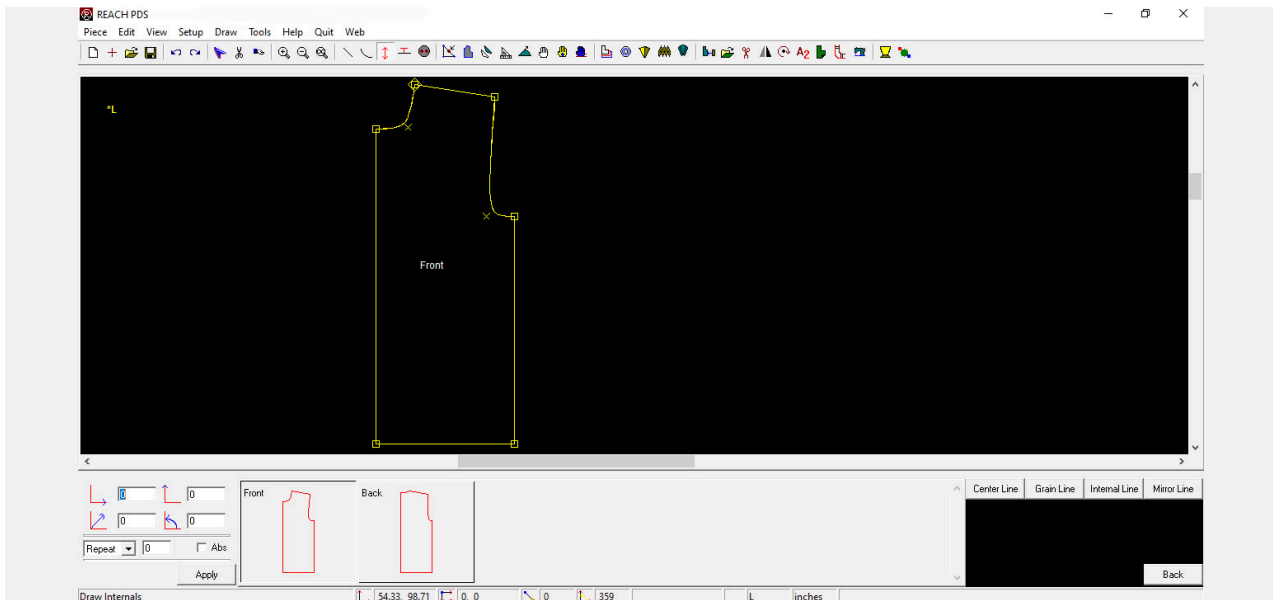


01

Understanding Internal Lines and Their Importance

1. Understanding Internal Lines and Their Importance

A. Internal lines are non-outline lines added to a pattern piece to provide guidance for placement, construction, symmetry, and sewing. They do not define the cutting edge but are critical for correct garment assembly and production alignment.

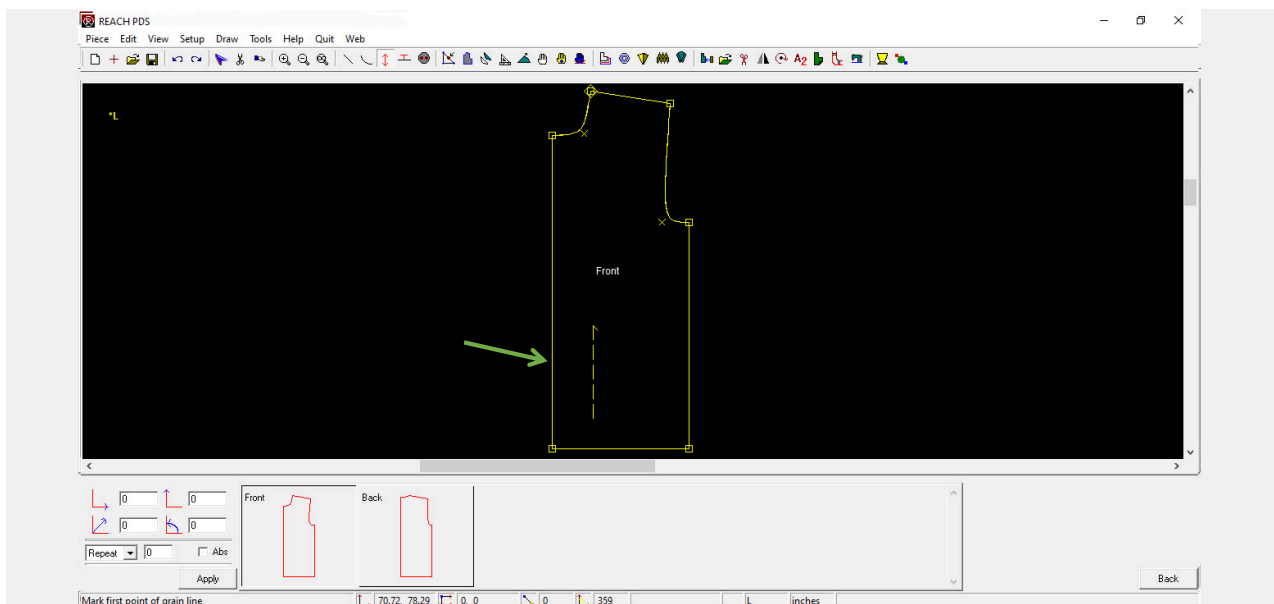
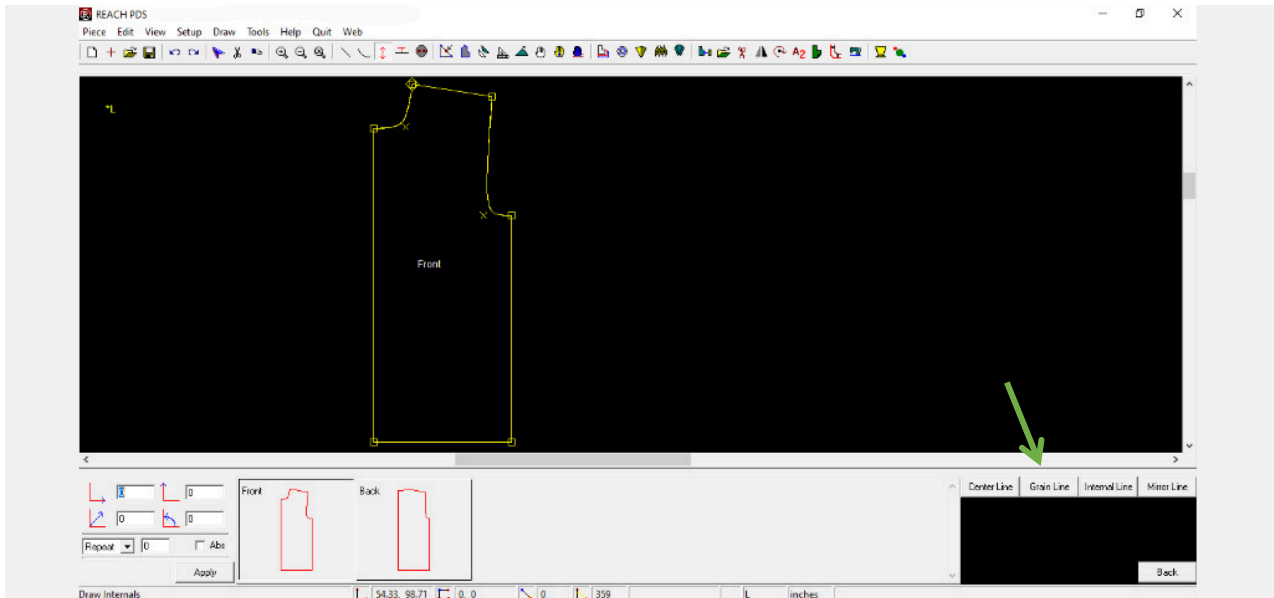


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B. Common Internal Lines:

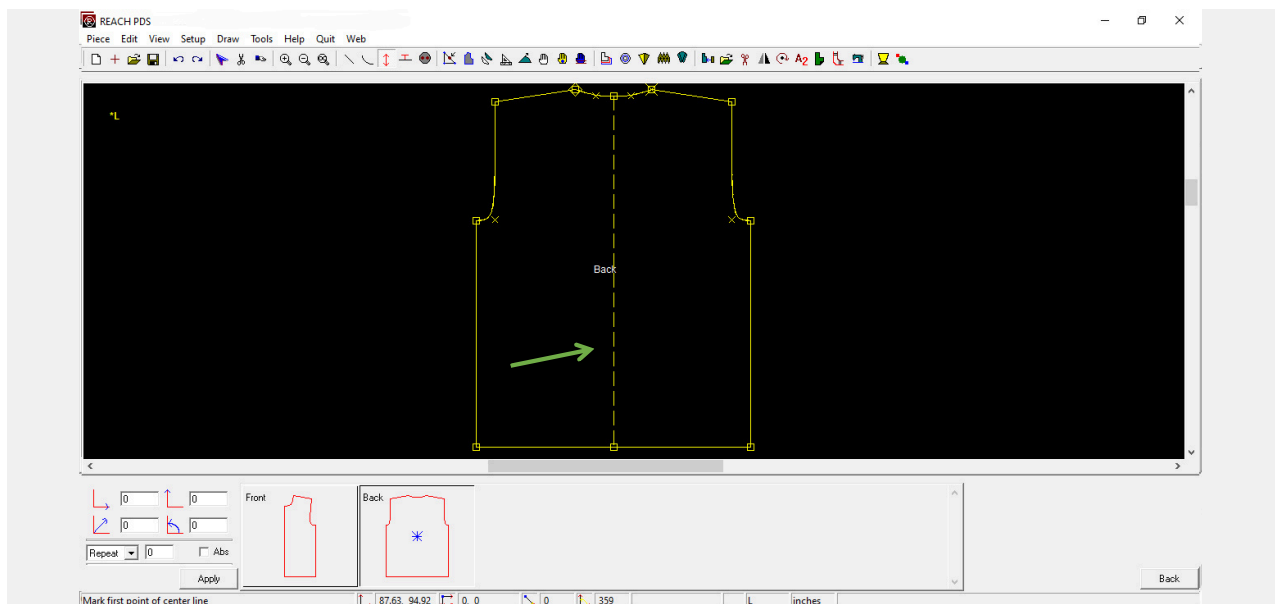
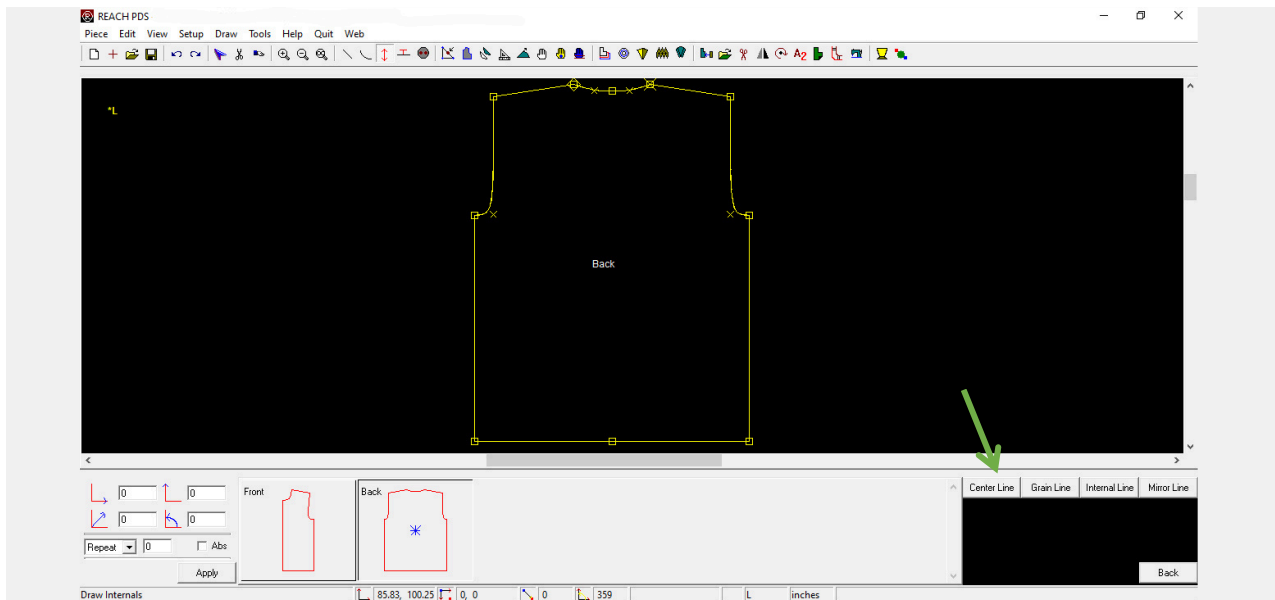
a. Grainline: Indicates the fabric grain direction for correct placement on fabric.



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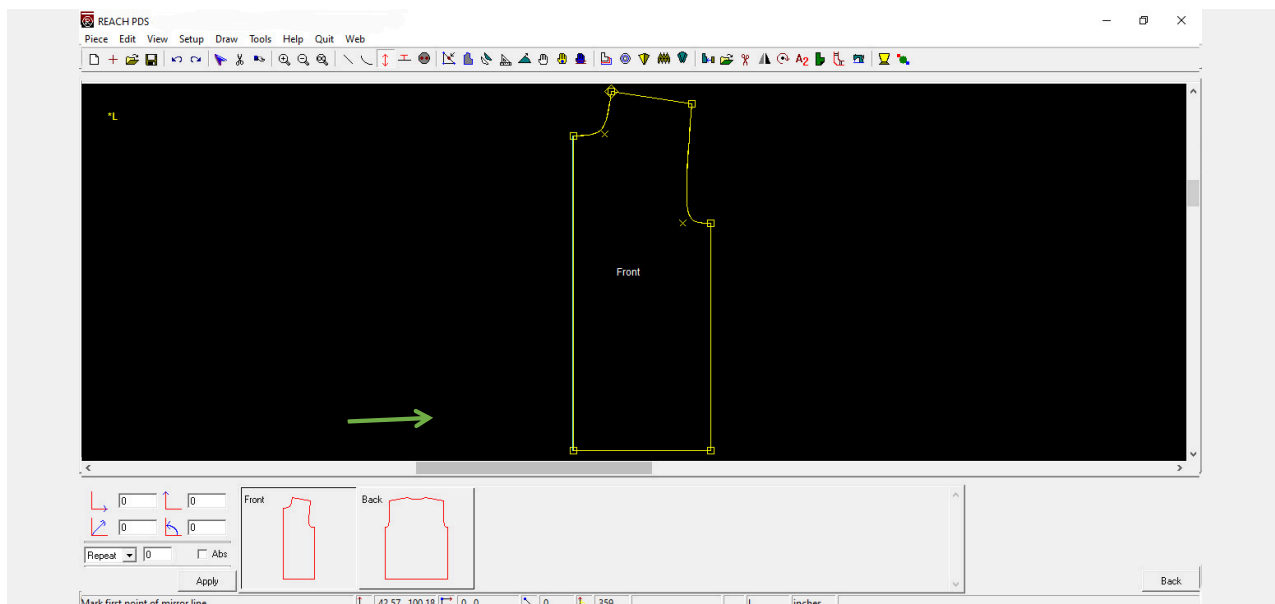
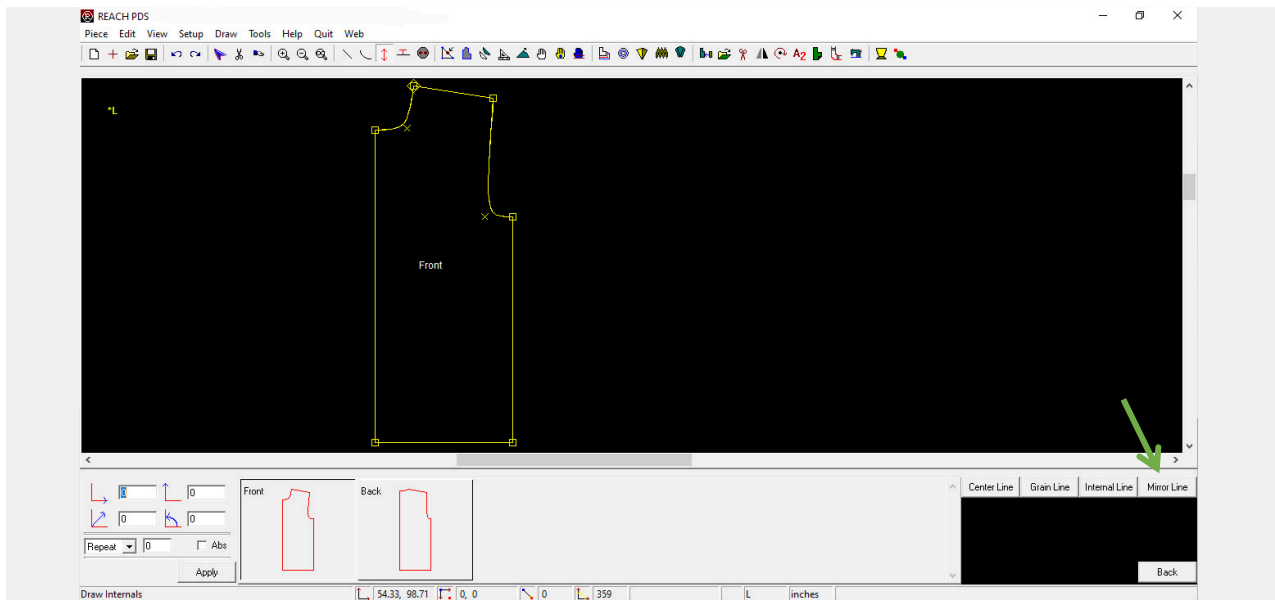
b. Center Line: Marks the center front or back, often used for alignment and folding.



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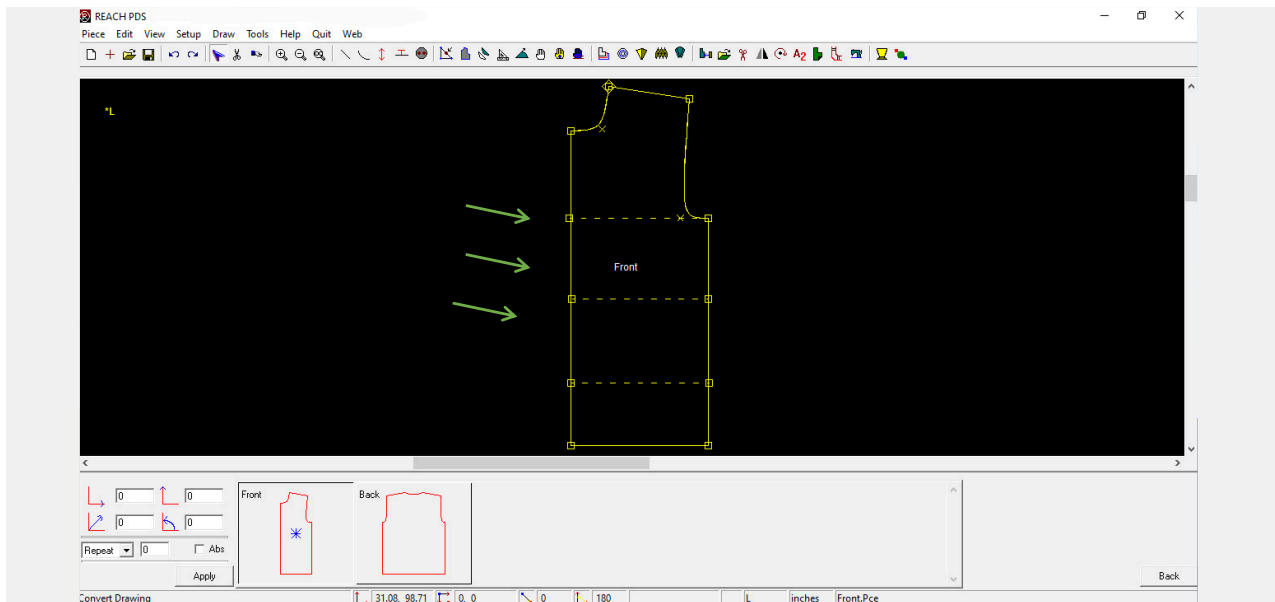
c. Mirror Line: Used for symmetrical designs or to indicate a fold.



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d. Balance Line: Helps with aligning pieces and checking symmetry, often drawn at bust, waist, or hip levels.



02

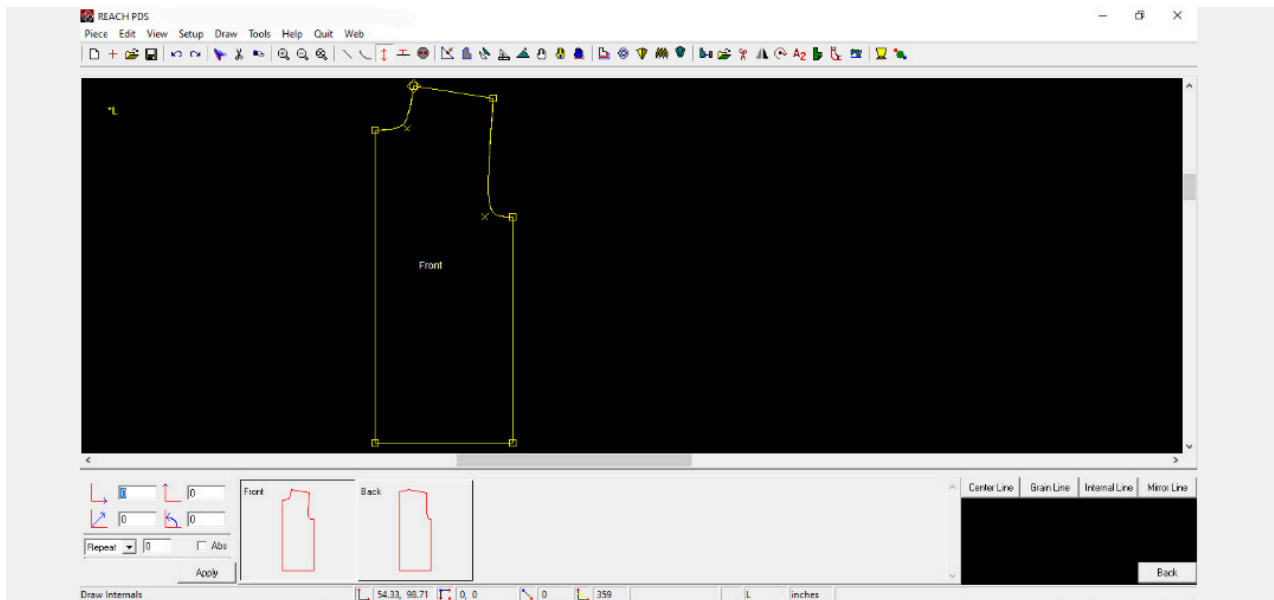
Adding Internal Lines in REACH CAD

2. Adding Internal Lines in REACH CAD

A. Adding a Grainline

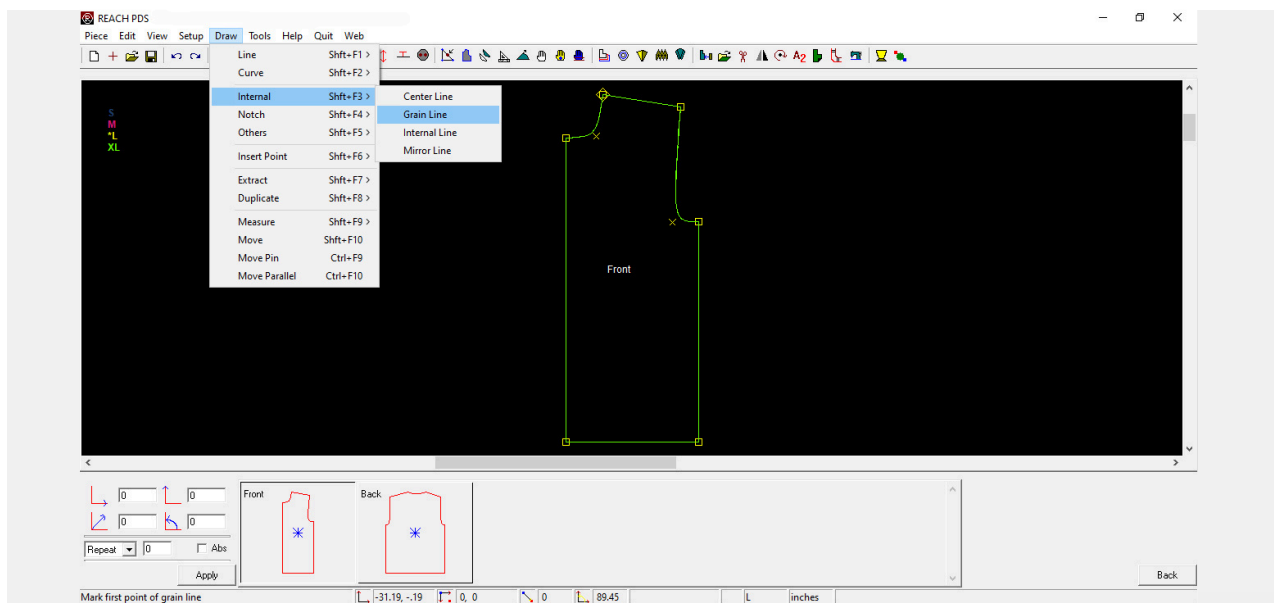
a. Select the Pattern Piece:

- i. Click on the piece (e.g., "Front_Bodice") you want to add a grainline to.



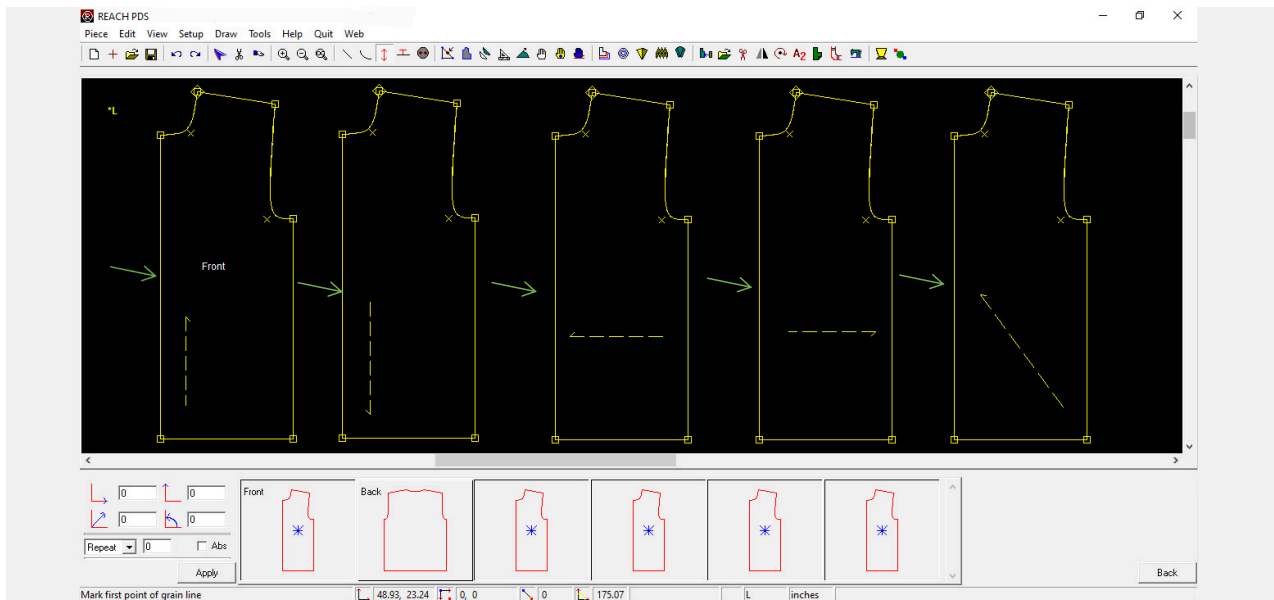
b. Draw Menu:

- i. Go to Draw > Internals > Grain Line.



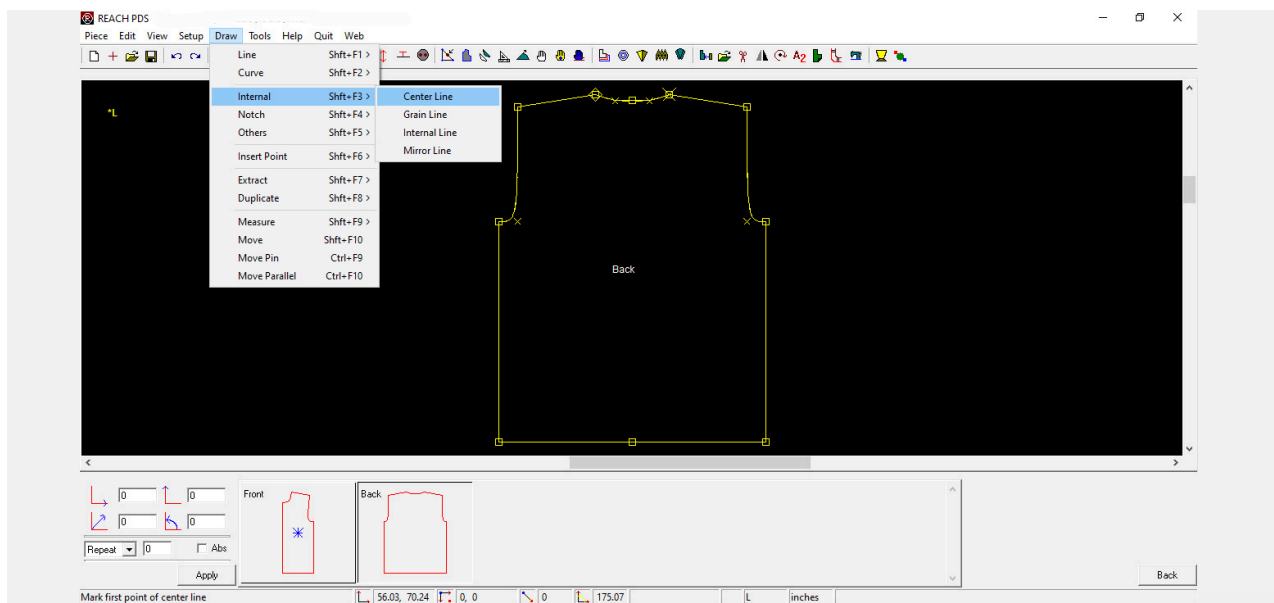
c. Placement:

- Click to place the grainline, usually parallel to the center front or back.
- Adjust its position as needed for correct fabric alignment.

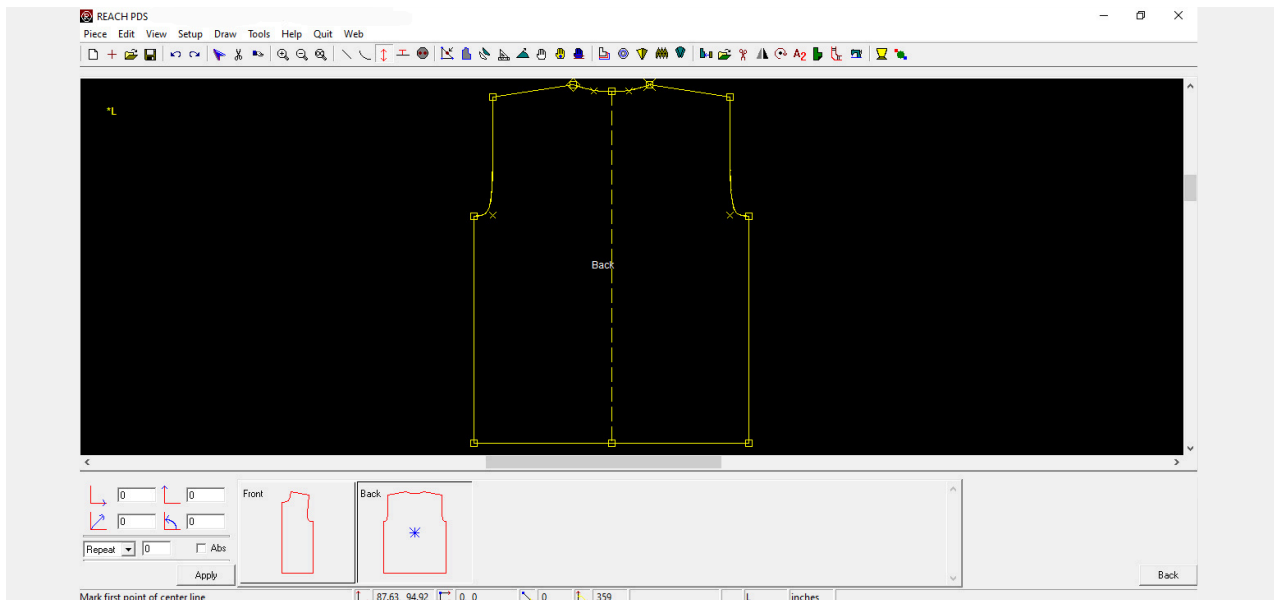


B. Adding a Center Line

a. Go to Draw > Internals > Center Line.

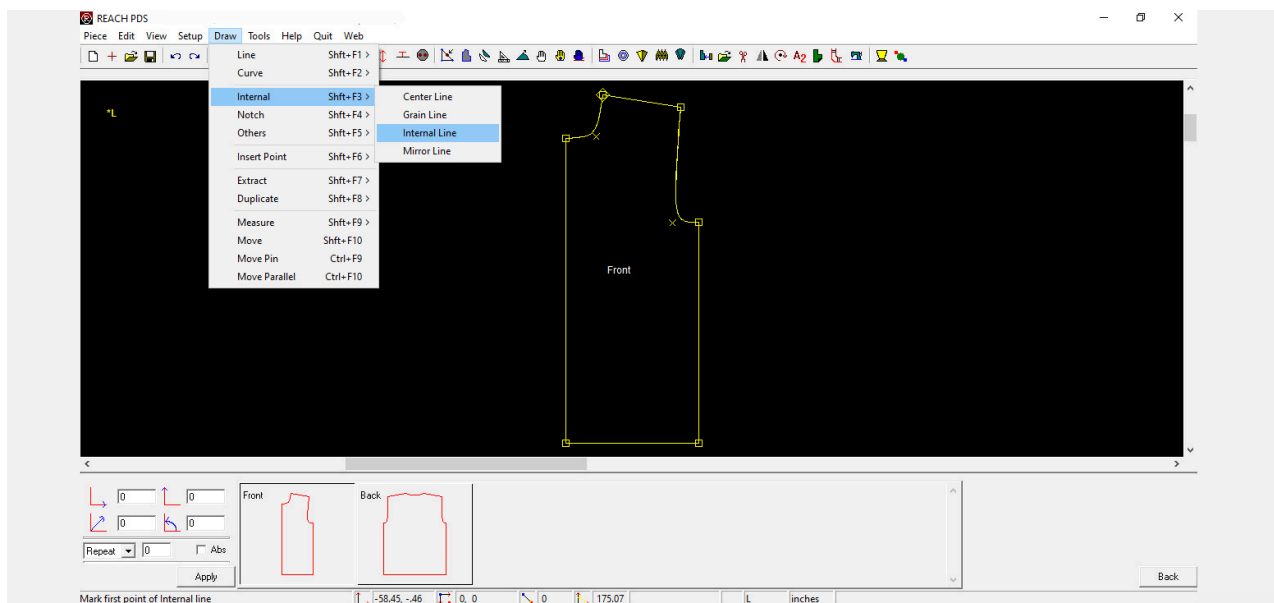


b. Place the line along the center of the pattern piece (e.g., center front or back).

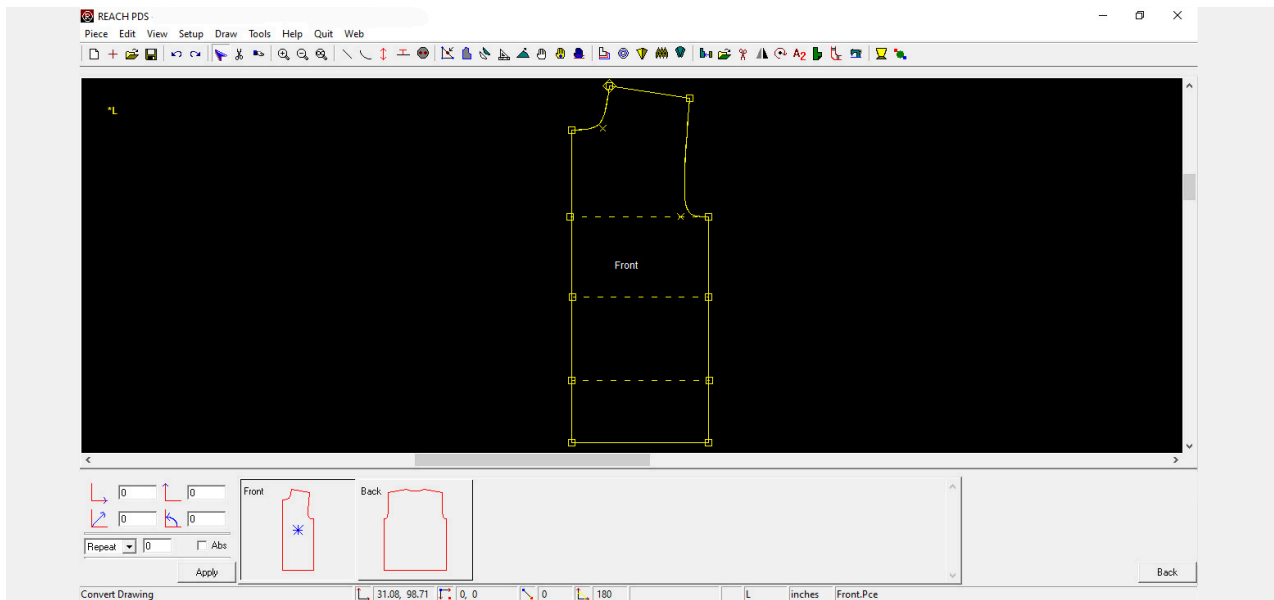


C. Adding a Balance Line

a. Go to Draw > Internals > Balance Line.

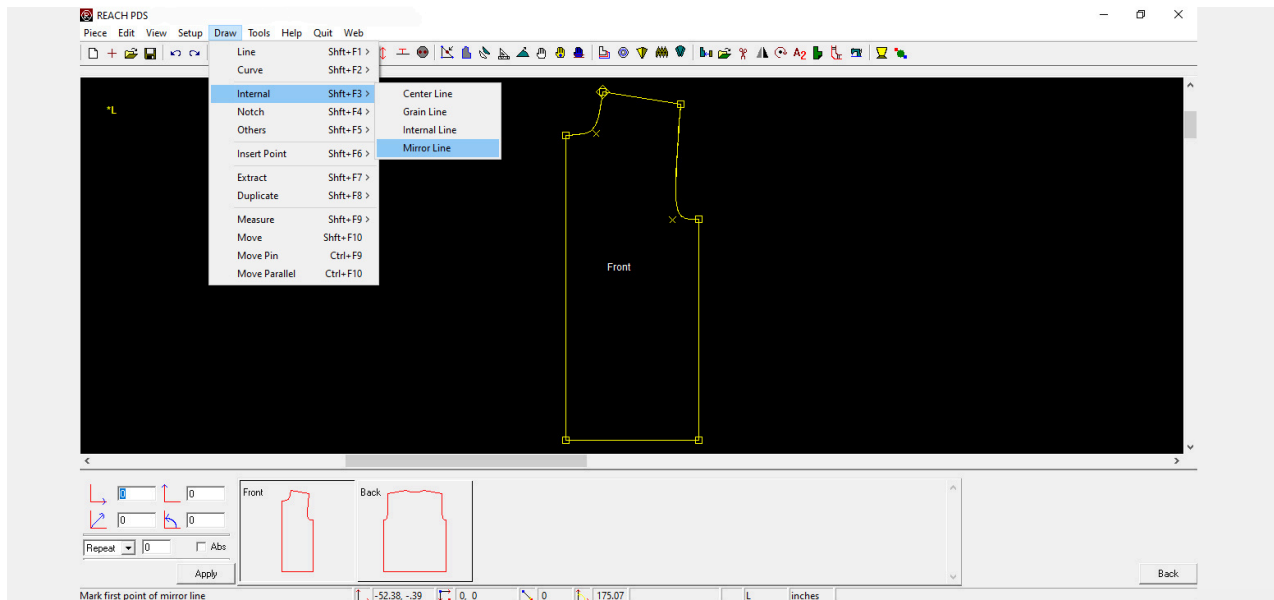


b. Draw a horizontal line at a strategic point (e.g., bust, waist, or hip level) for reference.



D. Adding a Mirror Line

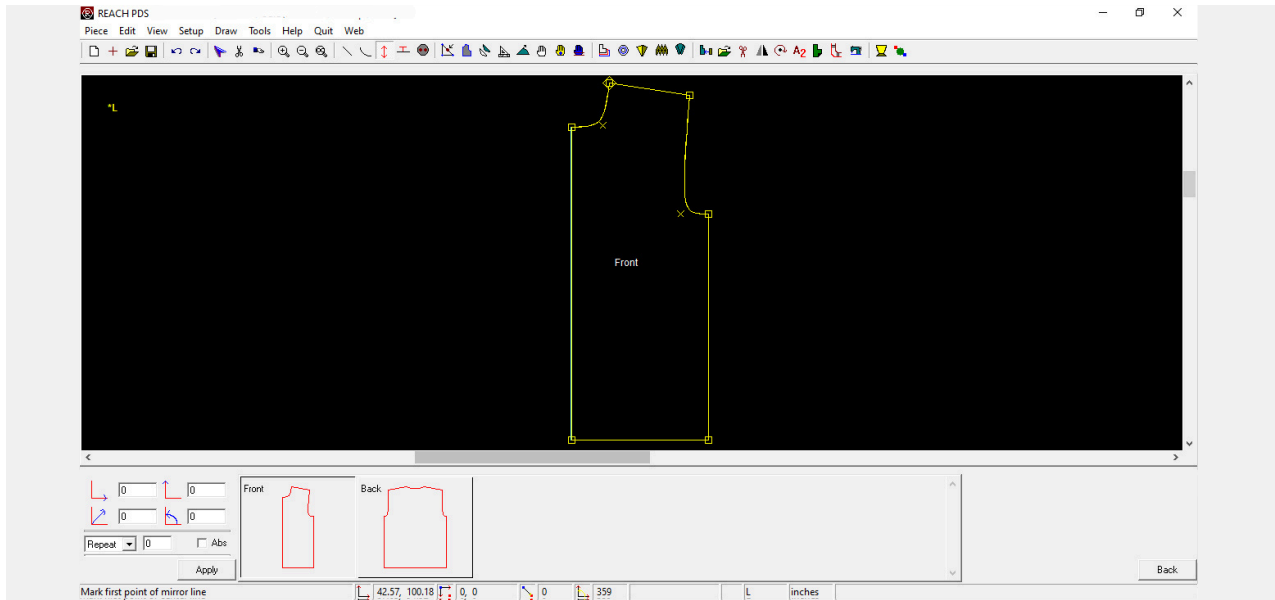
a. Go to Draw > Internals > Mirror Line.



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b. Place the line where the piece is to be folded or mirrored (often at center front or back).



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03

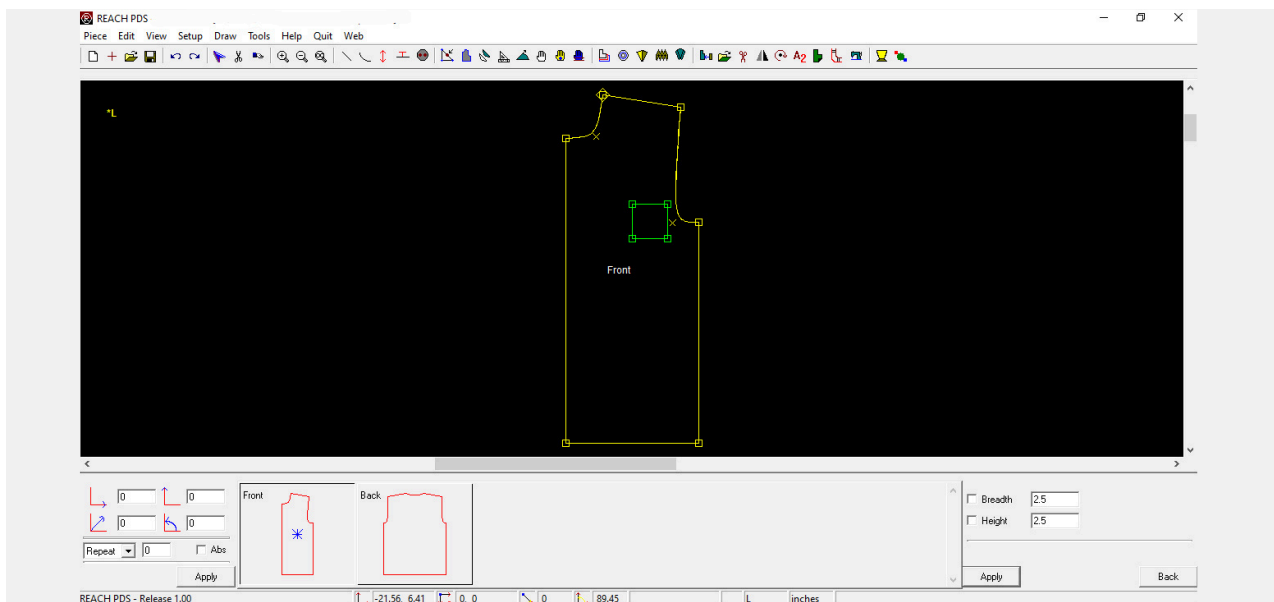
Using the Convert Tool

3. Using the Convert Tool

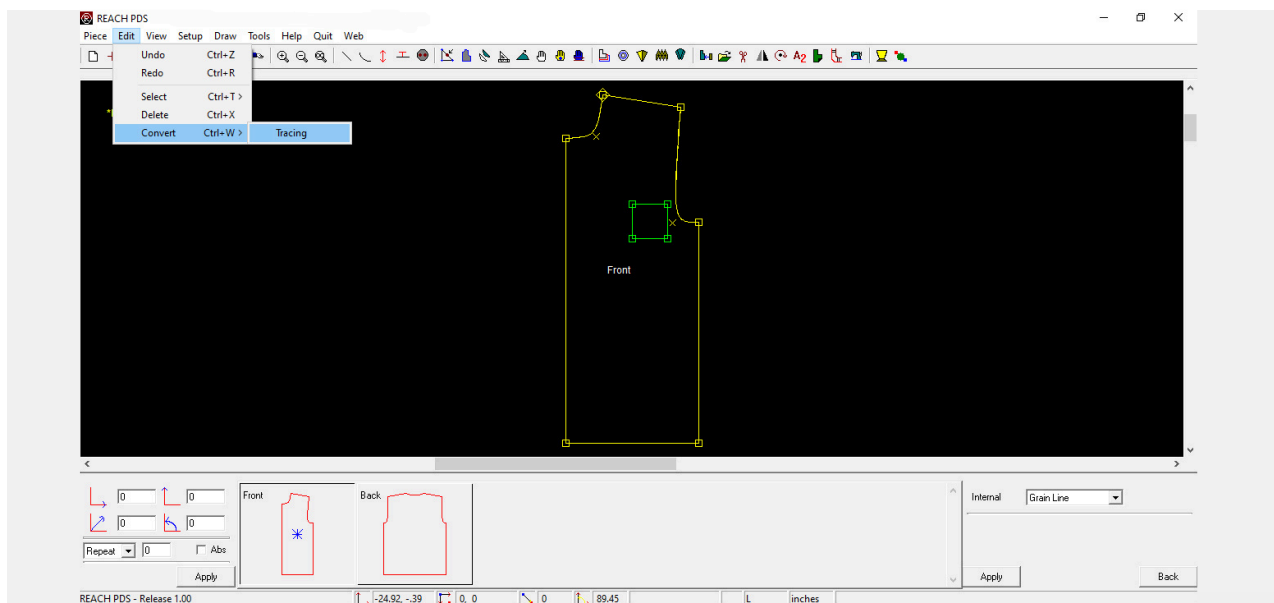
A. Purpose: The Convert tool in the Edit menu changes the property of a line or object (e.g., from an outline to an internal line or vice versa).

B. How to Use:

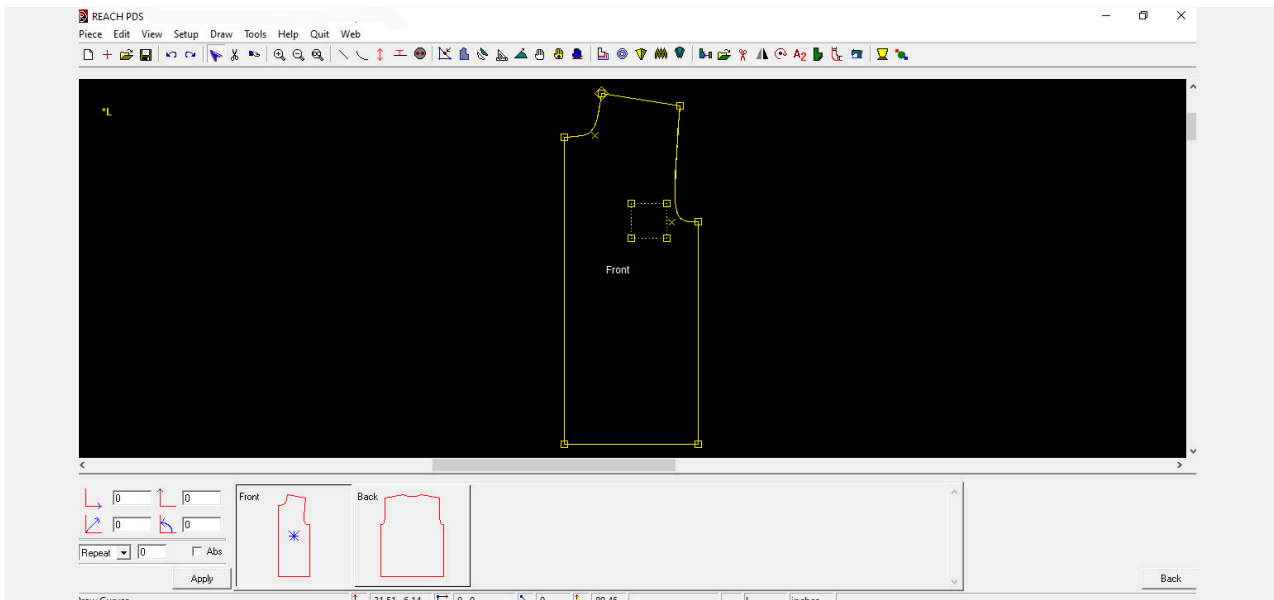
a. Select the line you want to convert.



b. Go to Edit > Convert.



c. Choose the desired property (e.g., convert an outline to an internal line).



C. Example: If you drew a construction line as an outline but want it as an internal sewing guide, use Convert to change its type.

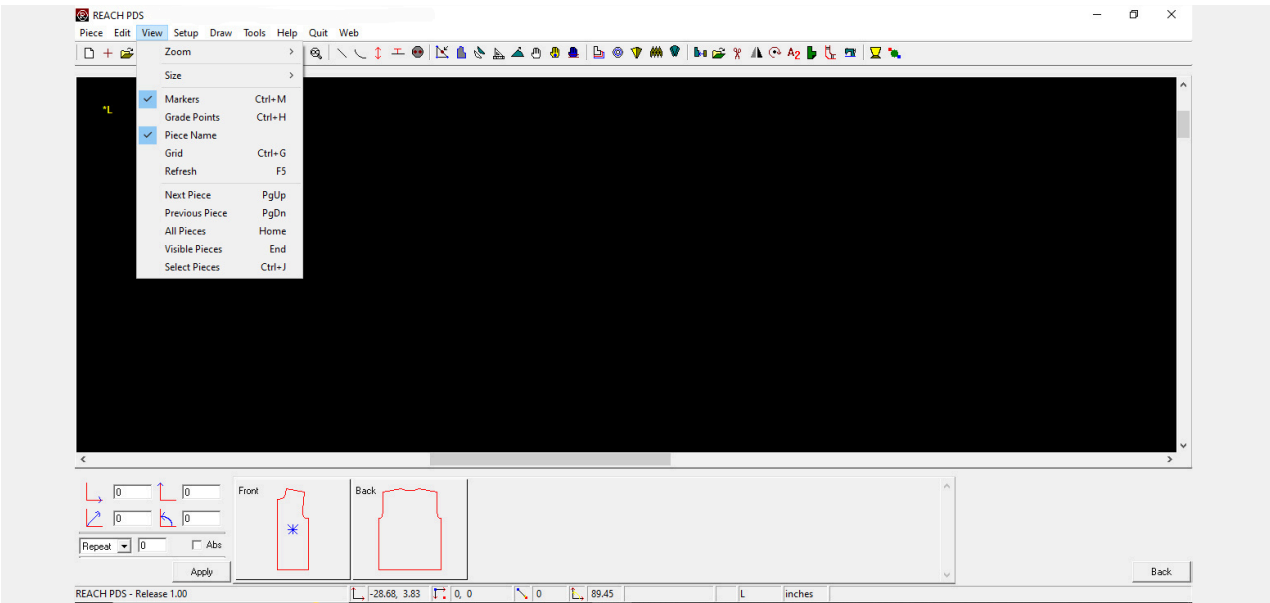


04

Advanced Viewing with the View Menu

4. Advanced Viewing with the View Menu

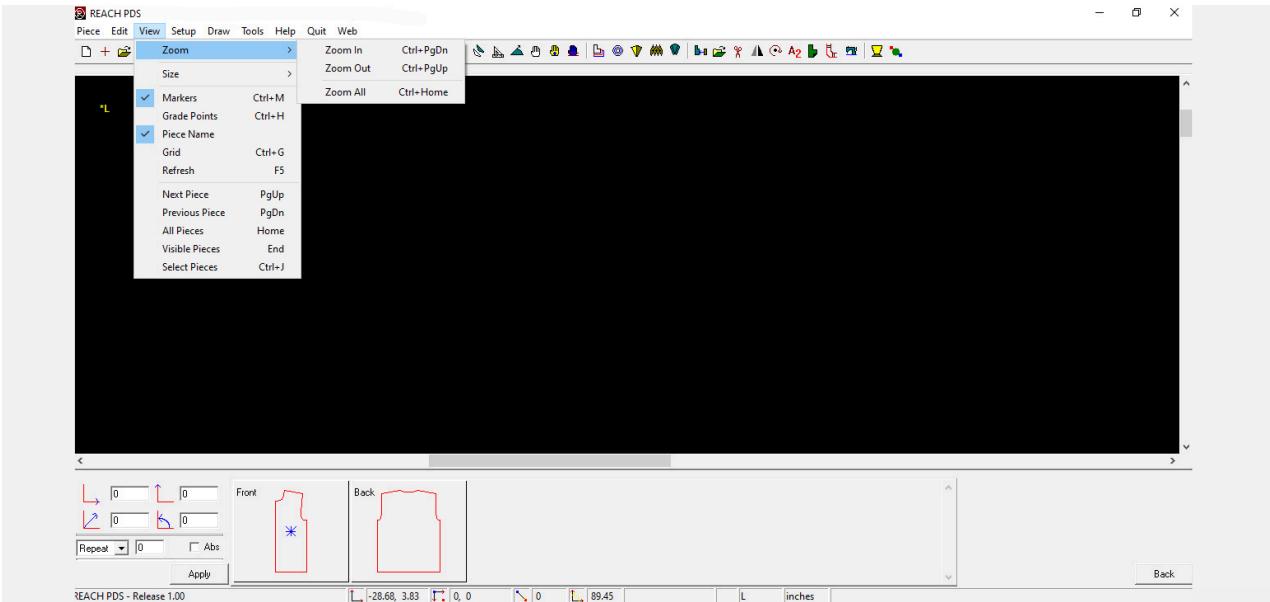
The View menu in REACH CAD allows you to control what you see on your workspace, making it easier to focus, inspect, and manage complex patterns.



A. Zoom and Navigation

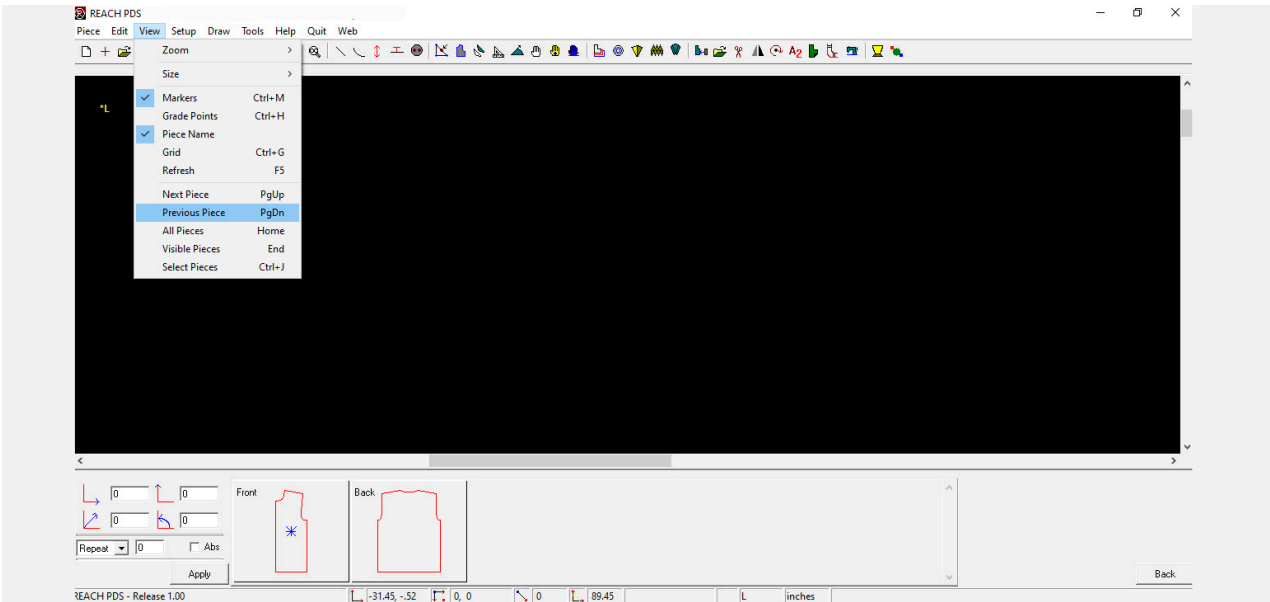
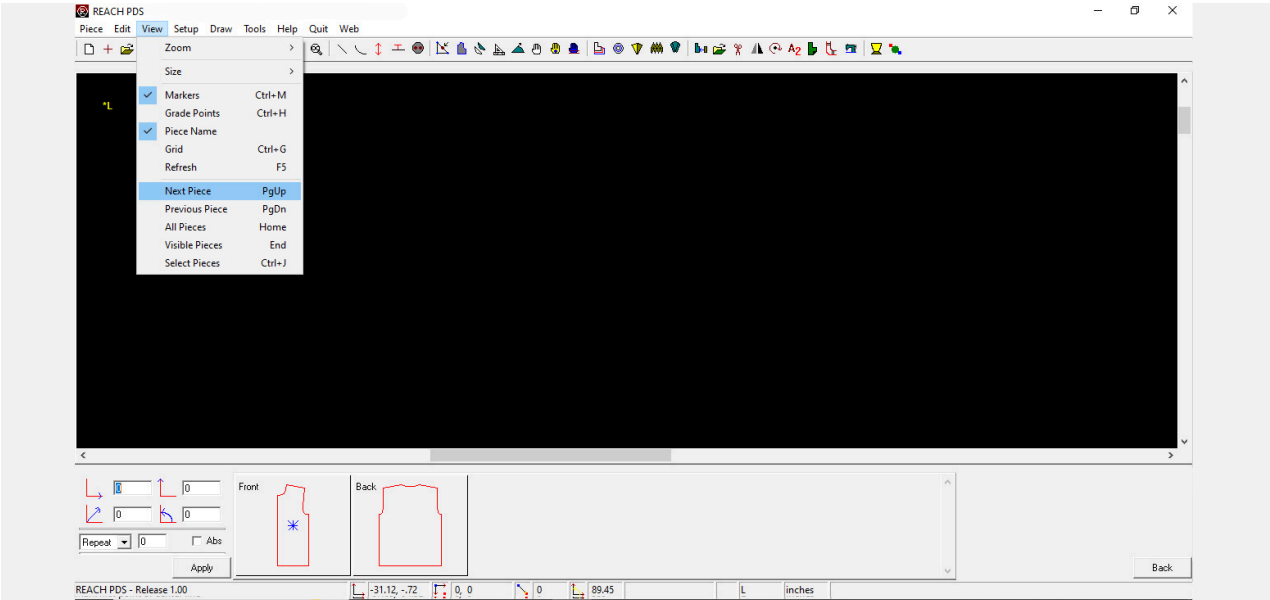
a. Zoom:

Use Zoom In/Out to closely inspect details (e.g., neckline curves) or get an overview of the entire piece.



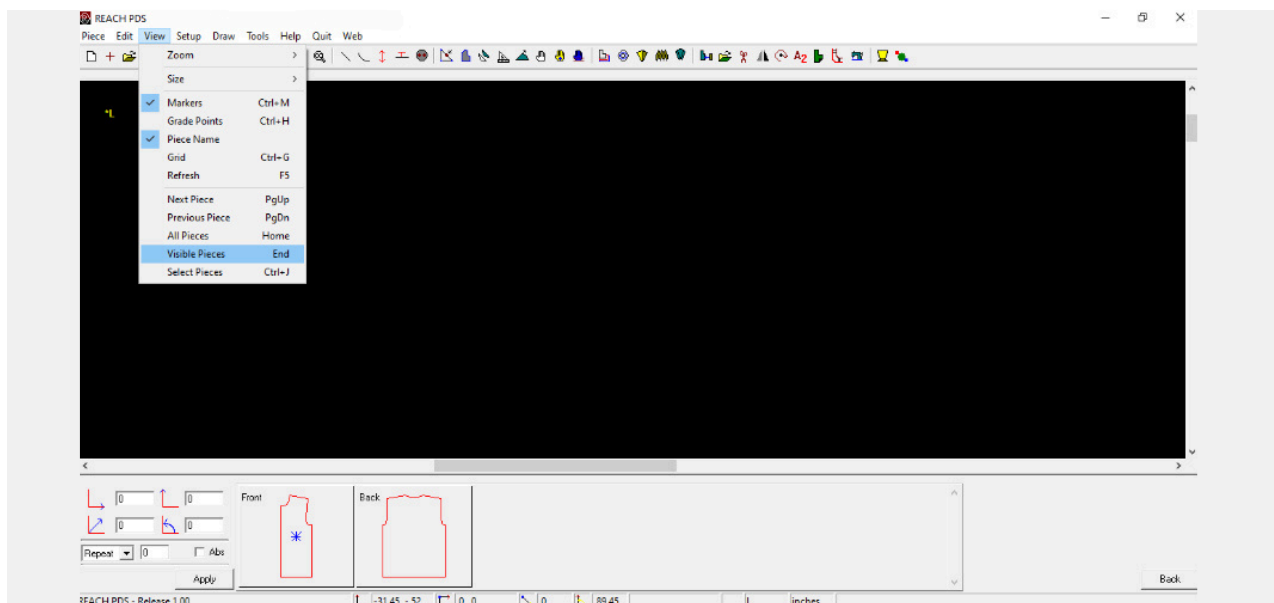
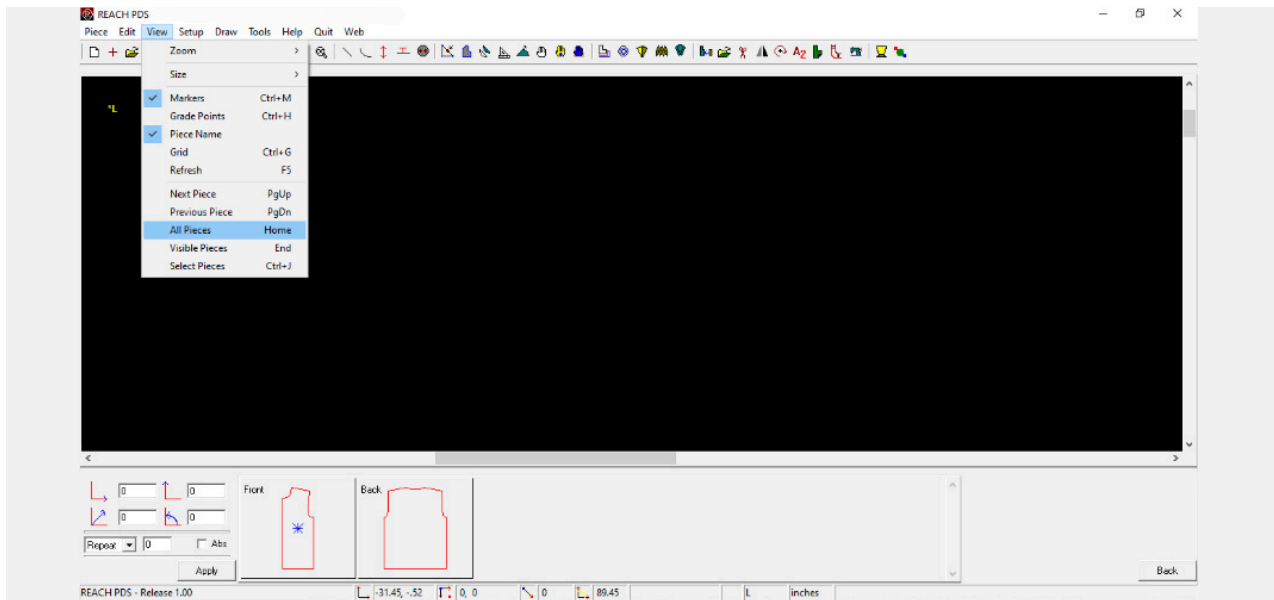
b. Next/Previous Piece:

Cycle through pieces in your style file one at a time.



c. All Pieces/Visible Pieces:

Show all pieces at once or only selected pieces for focused editing.



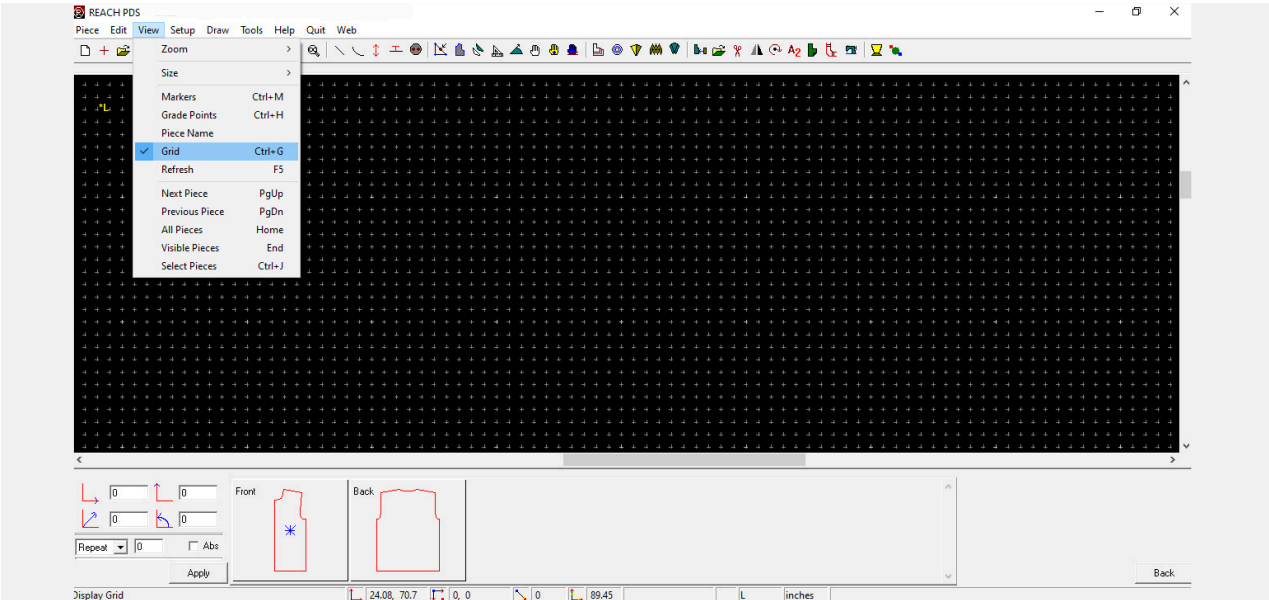
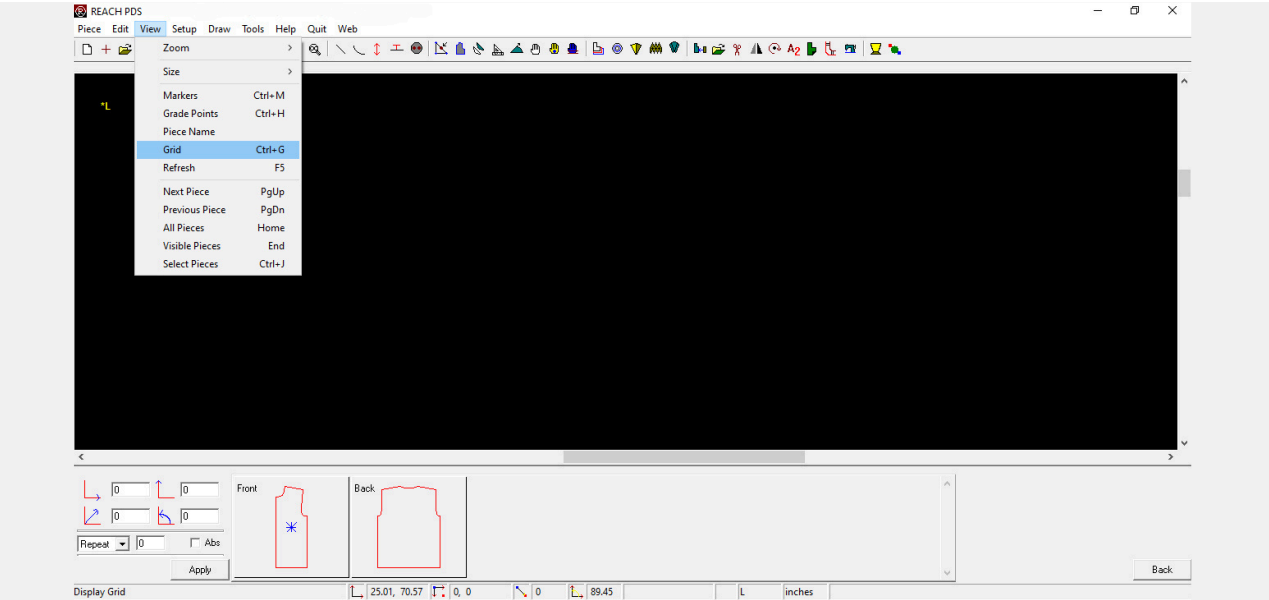
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B. Display Controls

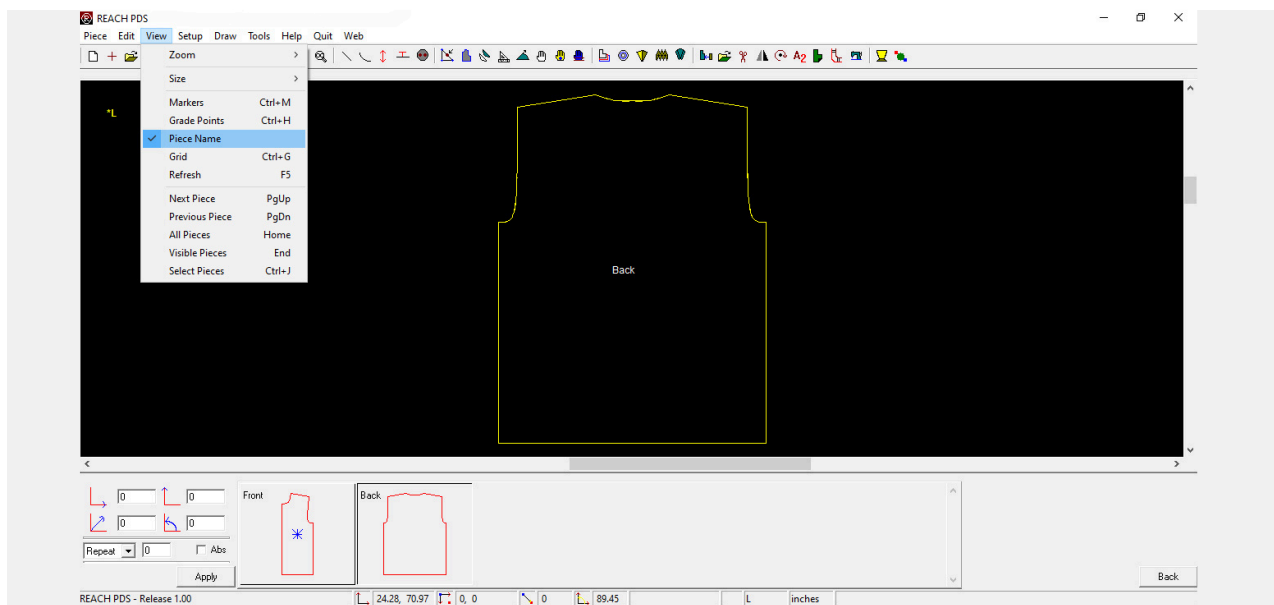
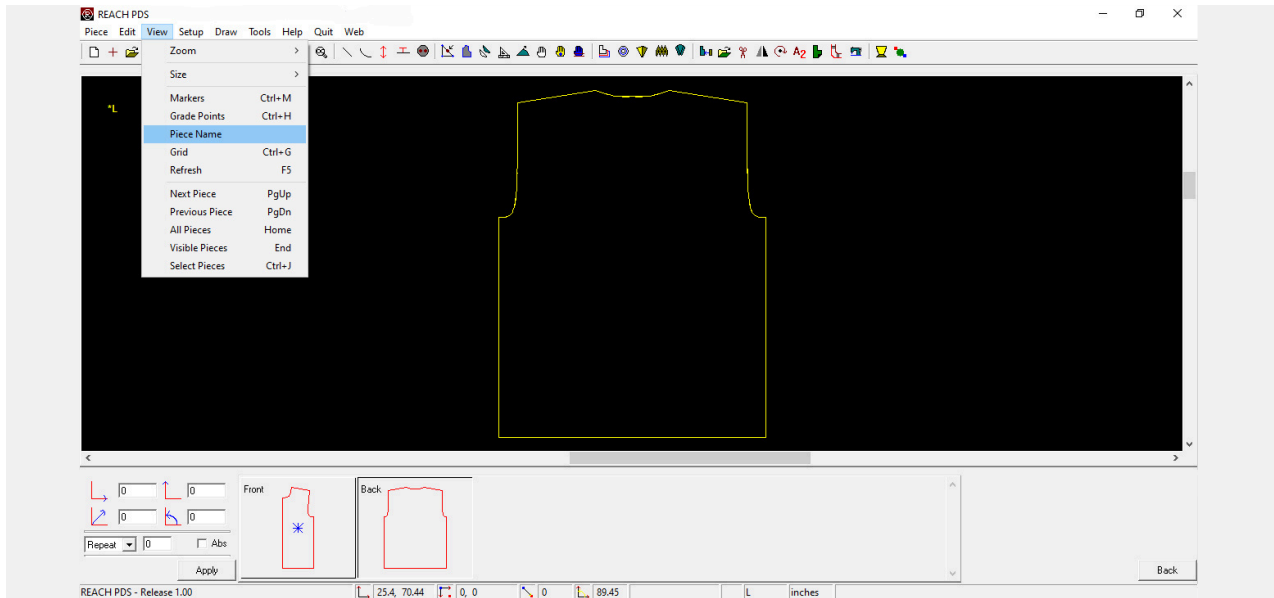
a. Grid:

Toggle the grid on/off for precise alignment of lines and points.

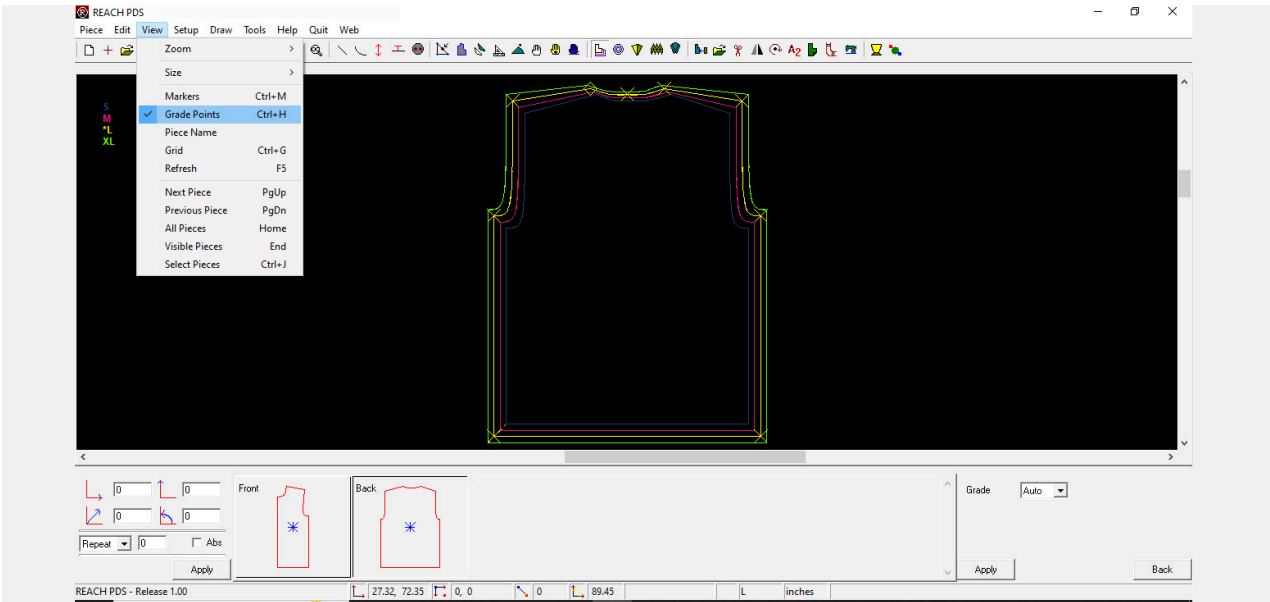
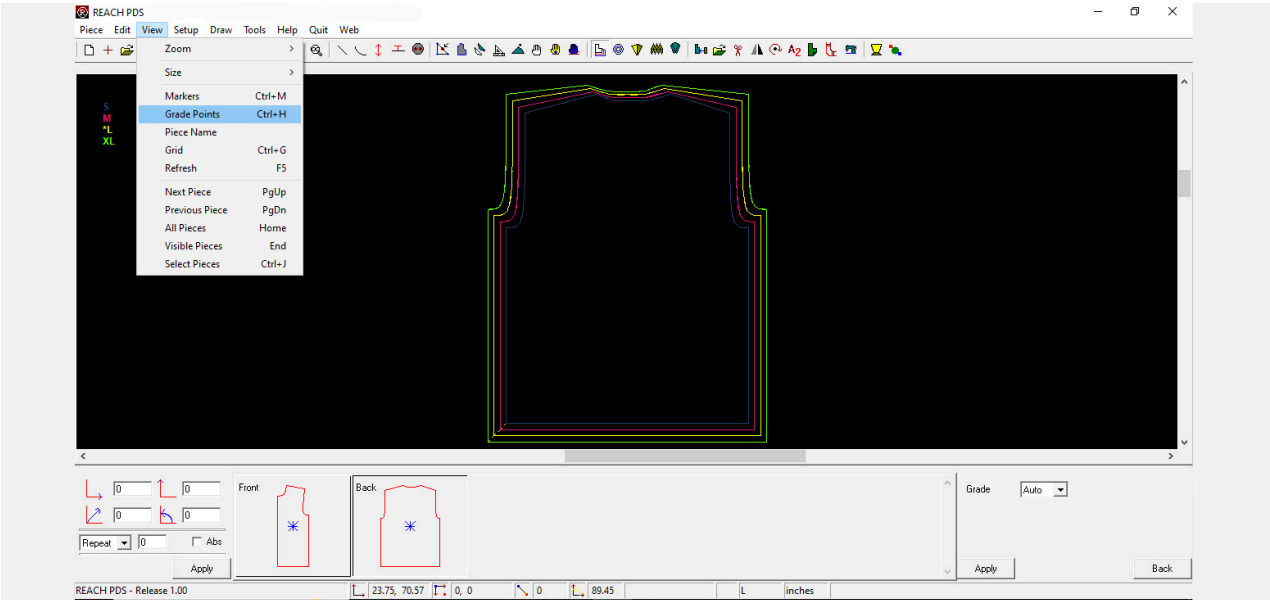


b. Show/Hide Elements:

Control the visibility of internal lines, piece names, grade points, or markers to declutter your workspace.



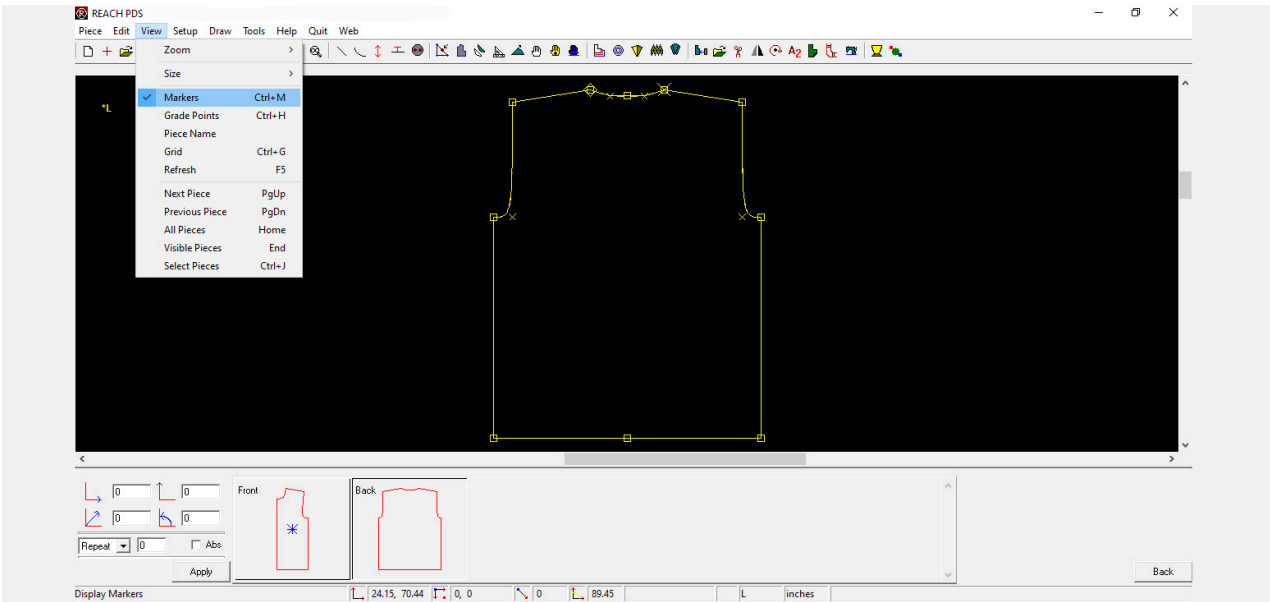
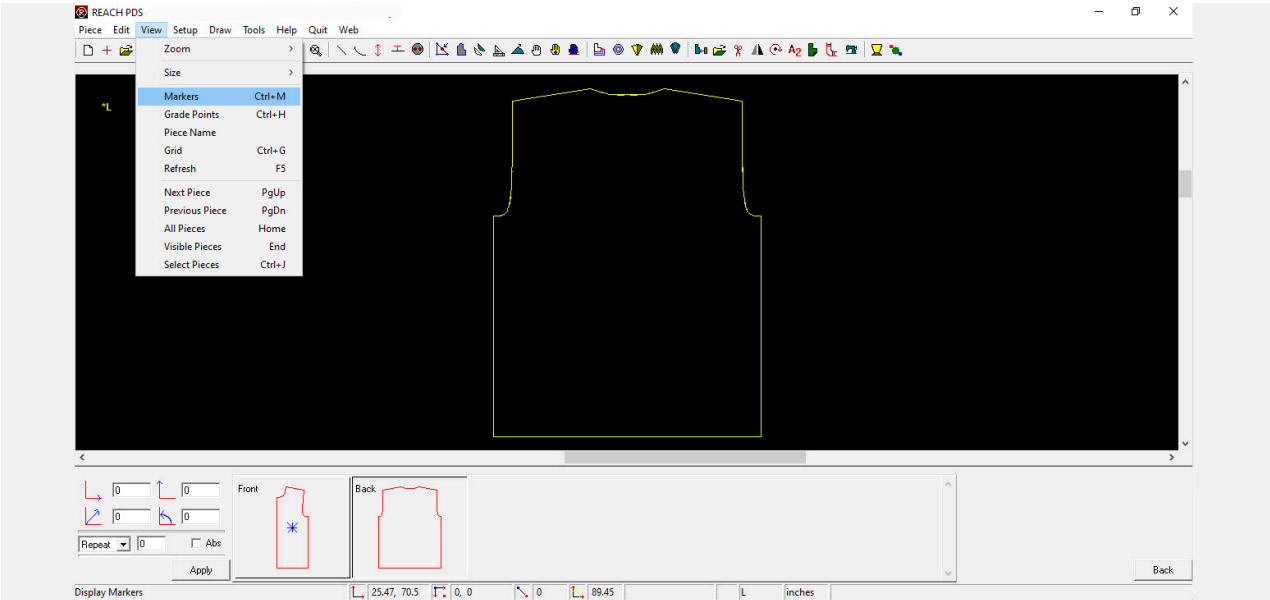
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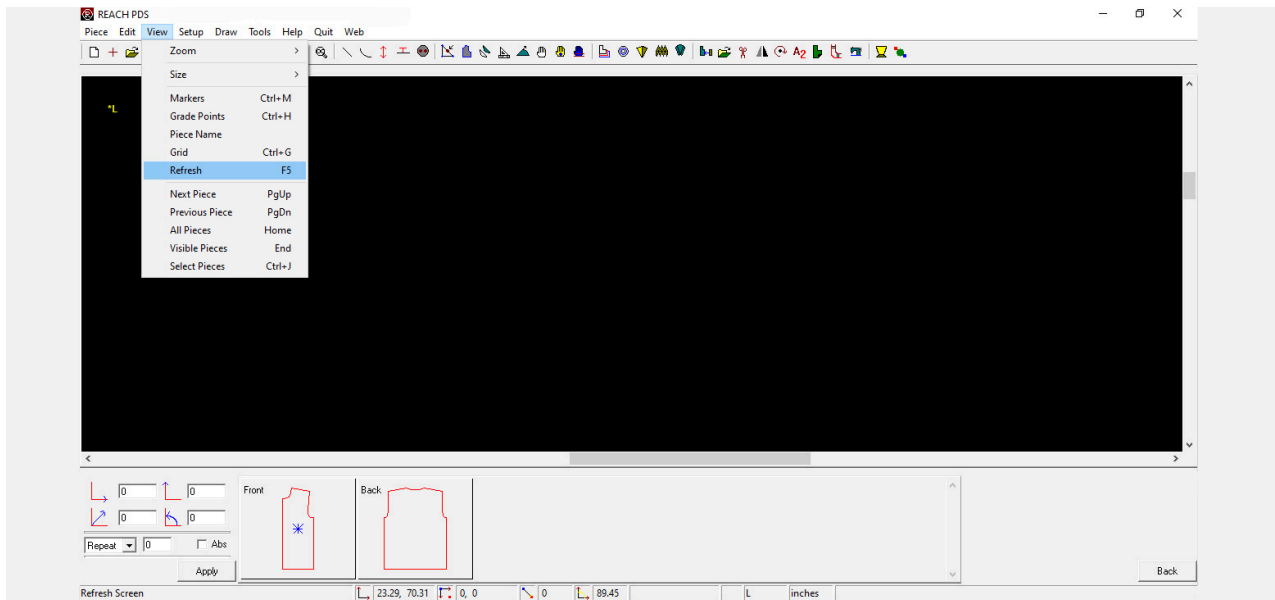


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c. Refresh

Use Refresh to update the screen after making changes, ensuring your display is current.



05

Practical Assignment

5. Practical Assignment

Try this:

- A. Open your Bodice Block style file.
- B. Select the "Front_Bodice" piece and add a grainline parallel to the center front.
- C. Add a center line and a horizontal balance line at bust level.
- D. Use the Mirror Line tool to create a fold line at the center front.
- E. Draw a standard line, then use the Convert tool to make it an internal line.
- F. Practice using the View menu to:
Zoom in on the neckline,
Show only the piece name,
Hide all internal lines,
Restore the full view.



06

MCQs and Short Answer Questions

6. MCQs and Short Answer Questions

Sample Multiple Choice:

1. Which internal line is crucial for placing the pattern correctly on the fabric?

- a) Center line b) Grain line **[Correct]** c) Mirror line d) Apex line

2. The Convert tool in the Edit menu is used to:

- a) Change the size of a piece b) Change the property of an object (e.g., from an outline to an internal line) **[Correct]**
- c) Change the unit of measurement
- d) Change the piece name

3. To see a magnified view of a specific area of your pattern, which View menu tool would you use?

- a) Zoom **[Correct]**
- b) Marker
- c) Size
- d) Piece name

4. What is the function of a 'Mirror line'?

- a) It defines the edge of the pattern.
- b) It indicates where to fold the fabric.
- c) It helps in creating symmetrical internal details. **[Correct]**
- d) It is another name for the grain line.

5. To hide all internal lines to see the outline of a piece clearly, you would use options within the:

- a) Setup menu
- b) Edit menu
- c) View menu **[Correct]**

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d) Draw menu

Sample Short Answer:

- A. Explain the purpose of adding a 'Grain line' to a pattern piece.
- B. What is the difference between an 'Internal line' and a piece's 'Outline'?
- C. Describe a scenario where the Convert tool would be useful.
- D. How do you delete an internal line without deleting the entire pattern piece?
- E. List three elements you can control (show/hide) using the View menu.



07

Best Practices

7. Best Practices

- A. Always add a grainline to every pattern piece before saving or exporting.
- B. Use internal lines for construction and symmetry guidance; keep outlines reserved for cutting edges.
- C. Use the Convert tool to fix any lines drawn with the wrong property.
- D. Master View menu controls to efficiently navigate and inspect complex patterns.



08

Key Takeaways

8. Key Takeaways

- A. Internal lines (grainline, center, mirror, balance) are essential for accurate pattern placement and construction.
- B. The Convert tool allows you to change any line's property for correct classification.
- C. The View menu is your control center for workspace clarity, navigation, and efficient editing.



In summary:

Adding internal details and mastering the View menu in REACH CAD are critical skills for professional pattern making. These tools ensure your patterns are production-ready, easy to interpret, and efficient to manage at every stage of the workflow.



enabling agile apparel enterprises for the digital economy

